Collation of Multilingual Versions of a Text: Necessity, Approach, Challenges

Balck, Sandra
sandra.balck@fu-berlin.de
Freie Universität Berlin, Germany

Dähne, Janis
janis.daehne@informatik.uni-halle.de
Martin Luther University Halle-Wittenberg, Germany

Etling, Fabian
fabian.etling@fu-berlin.de
Freie Universität Berlin, Germany

Fischer, Frank
fr.fischer@fu-berlin.de
Freie Universität Berlin, Germany

Frenzel, Steffen
steffen.frenzel@uni-potsdam.de
Universität Potsdam, Germany

Grote, Brigitte
brigitte.grote@fu-berlin.de
Freie Universität Berlin, Germany

Hesse, Sascha
sascha.hesse@informatik.uni-halle.de
Martin Luther University Halle-Wittenberg, Germany

Molitor, Paul
paul.molitor@informatik.uni-halle.de
Martin Luther University Halle-Wittenberg, Germany

Pöckelmann, Marcus
marcus.poeckelmann@informatik.uni-halle.de
Martin Luther University Halle-Wittenberg, Germany

Ritter, Jörg
ritter@informatik.uni-halle.de
Martin Luther University Halle-Wittenberg, Germany

Singh, Yashee
yashee.singh@fu-berlin.de
Freie Universität Berlin, Germany

Stede, Manfred
stede@uni-potsdam.de
Universität Potsdam, Germany

Introduction

Texts written in different languages, which are not literal translations from a source language to a target language, but rather multilingual versions of a text, are the subject of research in numerous disciplines. These multilingual text versions can be found among authors who write in two or more languages and for whom switching between languages always means adapting the text to a new target group. In literary works, the translation into another language is often accompanied by targeted interventions in the source text, e.g., through extensive additions, deletions and rephrasings. The following contribution deals with cross-linguistic alignment and presents an integral approach to this central problem in literary studies.

The COMUTE-Project

The project ‘Collation of Multilingual Text’ (COMUTE), a research project funded by the German Research Foundation (DFG)[1], develops and implements an e-research technology for the comparison of such variant text versions. These versions are neither literal nor sense-for-sense translations from a source language into a target language. To approach these complex multilingual versions we extend the research software LERA[2], developed at the Martin Luther University Halle-Wittenberg. LERA is currently used in a large number of editing projects[3] for the collation of text witnesses written in the same language (see Fig. 1).

Use Cases

One prominent example for multilingual versions of text is Hannah Arendt's work, which will also serve as a first use case for the COMUTE

project. Arendt wrote her major post-war books (Origins of Totalitarianism / Elemente und Ursprünge totaler Herrschaft; Human Condition / Vita activa; On Revolution / Über die Revolution) first in English and then in German. The versions differ considerably: According to an initial estimate, only approx. 20% of Human Condition can be described as a literal translation. A further 20% of the text only appears in one language, while the remaining 60% shows similarities between the linguistic versions, but are not literal translations.

Other examples are Heinrich Böll's adapted German translation of Salinger's novel The Catcher in the Rye; and the numerous German translations of Hermann Melville's Moby Dick and the associated debate about their different political meanings and readings.[4]

This complexity can also be found in the social sciences. For instance, a number of constitutions have been used as key examples for those of other countries (especially in the area of fundamental rights). The constitutions of the USA and Switzerland, as well as the German Basic Law, for example, served as a source of inspiration for the constitutions of Eastern European countries that emerged in the 1990s (Knoth et al. 2018).

State of the Art

The collation of multilingual versions of only partially parallel texts is of interest to all academic fields in which texts are either the subject of investigation or represent a central medium for securing academically relevant information. Although there have been a number of studies comparing multilingual versions of a text, as for example VecAlign (Thompson / Koehn 2019), SentAlign (Steingrímmsson et al. 2023), or CroCoAlign (Molfese et al. 2024), they are all limited to comparing exactly two texts that are largely literal translations of each other and only consider sentence alignment. There is no detailed comparison at the level of words or of elementary discourse units (see below). Furthermore, there is currently no (user-friendly) working environment for comparing multilingual versions of a text.

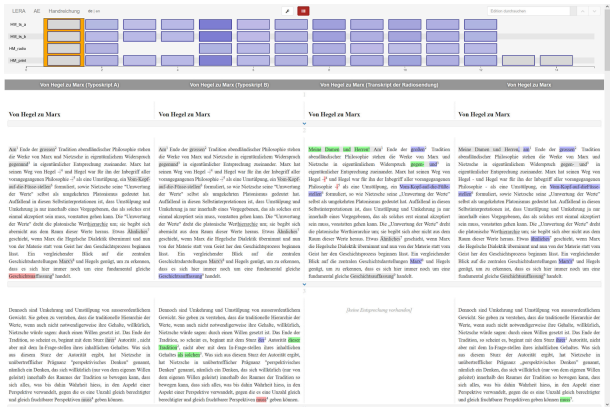

Fig 1. Screenshot of LERA applied to same-language versions of a text.

Objective

In order to support researchers as much as possible, COMUTE develops a paragraph and sentence alignment with a detailed comparison of the aligned paragraphs or sentences. Distance measures like Jaccard similarity (Jaccard 1901) or Levenshtein distance (Levenshtein 1966), which are used in current projects to compare monolingual text versions, e.g., by CollateX (Dekker et al. 2014) or LERA, cannot be used for cross-lingual text comparisons. Word-to-word alignments are not really expedient, even when the text versions are literal translations of each other, as the word number and order can be different in contrasting languages. If the text versions to be compared are not literal translations of each other, they are even largely useless. Therefore, detailed comparison of aligned paragraphs must relate to the elementary train of thought contained in the sentences.

Process

For the comparison of multilingual versions, we will implement a three-step process. The first step is to perform an n-to-m alignment of the paragraphs or sentences (e.g., by VecAlign, SentAlign, or CroCoAlign). Next, we extract elementary discourse units (EDU) from the paragraphs for detailed analysis. In the final step, a matching of EDUs in aligned paragraphs is calculated and visualised. Hence, the detailed analysis is not a word-to-word-, but an EDU-to-EDU-alignment. This approach follows the premise that a sequence of EDUs constitutes a more fine-grained representation of a train of thought, and while sentences can be re-worked in a revision of a text, EDUs representing single events or ideas are more sta-

ble units for alignment across text versions. The notion of "Elementary Discourse Units" (EDUs) originated in the field of discourse parsing, especially in the tradition of Rhetorical Structure Theory (RST) (Mann / Thompson 1987), where a text is first divided into EDUs and then recursively connected to each other via coherence relations (Cause, Contrast, Elaboration, etc.). Intuitively, an EDU is an independent clause or an adjunct clause that makes a complete contribution to the discourse. Specific annotation guidelines then typically describe language-specific syntactic criteria.

An early EDU segmenter for English, built on top of a syntactic parser, was SLSeg (Tofiloski et al. 2009), and a similar German variant was presented by (Sidarenka et al. 2015). A more recent approach using a BiLSTM-CRF approach is NeuralEDUSeg (Wang et al. 2018). Recently, a few multilingual models have been built as part of a shared task (Braud et al. 2023). For our EDU segmentation of German data, we use a modified version of the DPLP RST parser (Ji / Eisenstein 2014), which was trained by Shahmohammadi and Stede (2024) on German blogpost data and the Potsdam Commentary Corpus (Stede / Neumann 2014). It produces RST trees, from which we retrieve the EDU-segments.

Challenges of Text Alignment

As part of the project, we are also addressing two issues that not only need to be solved in our scenario, but pose challenges in the context of monolingual text versions too.

The choice of the sentence alignment method, if based on a similarity measure such as cosine similarity of sentence embeddings[5], depends on whether two or more text versions are to be compared. In the case of two text versions, the alignment problem can be efficiently and optimally solved, regarding the similarity measure used, by calculating the most efficient path in an directed acyclic graph, via which the alignment problem can be modelled (Cormen et al. 2009). Sentence alignment for multiple text versions is much more complex and requires considerable computing time for 'optimal' results, so good heuristics are required here.

A further challenge arises when, as in our project, versions of a text have already undergone major revisions. If paragraphs have been split into two or more paragraphs with the aim to add a new thought within the revision, current sentence/paragraph alignment approaches lead to poor results. To solve this problem, LERA offers the option of manually resegmenting the texts at critical text passages. VecAlign and SentAlign[6] are the first to present approaches for calculating n-to-m alignments for two texts to be collated at the sentence level. Unfortunately, both VecAlign and SentAlign only recognise the situation in which a sentence is split into two or more smaller sentences. If a sentence has been split up to insert new thoughts, this often leads to poor alignments at these points.

We are currently working on an approach which, prior to calculating the paragraph alignment, resegments the two texts at paragraph level based on the given paragraph segmentation of the texts and a 1-to-1 alignment of the sentences, which is calculated independently of the paragraph segmentation. It is done in such a way that the subsequent calculation of the paragraph alignment delivers qualitatively better results than without this resegmentation.[7]

Final Remarks

In summary, we tackle the problem of aligning multilingual versions of text with an hierarchical approach on the paragraph-, sentence- and EDU-level. Therefore we are solving issues of current approaches that only allow the alignment of exactly two witnesses that ideally are close translations of each other and have not undergone any major resegmentation. Besides overcoming these limitations, we plan to extend the graphical user interface of LERA to ensure that the solutions are accessible to researchers. With the applications of EDUs, there is a promising approach to add a further layer of a detailed semantic collation that goes beyond sentence alignment. Our presentation will show the latest results and discuss further questions such as improvement and gerneralisation of the approaches presented.

Notes

1. As part of the Scientific Library Services and Information Systems (LIS) funding programme ( )
2. See (Pöckelmann et al. 2023) and https://lera.u-zi.uni-halle.de
3. There are currently around 90 applications of LERA: in particular, the project “Kalīla and Dimna: AnonymClassic and Arabic Literature Cosmopolitan” (https://kalila-and-dimna.fu-berlin.de), „Hannah Arendt. Critical Complete Edition” (https://hannah-arendt-edition.net), “HyperAzpilcueta - Visualising the instability of early modern normative knowledge” ( ), and “The Kabbalistic Treatise Keter Shem Tov: Synoptic Edi-

tion, Commentary, Translation and the History of its Reception" (https://kabbalaheditions.org).
4. See, e.g., the translation dispute between Matthias Jendis and Friedhelm Rathjen (Habeck 2002).
5. A numeric representation of a sentence in the form of a real numbered vector, as used by sentence transformer models, e.g. SBERT (Reimers / Gurevych 2019).
6. A similar approach developed at the same time as that of Steingrímsson et al. (2023) is presented in (Dähne / Ritter 2024).
7. Initial results can be found in (Dähne et al. 2025).

Bibliography

Braud, Chloé / Liu, Yang Janet / Metheniti, Eleni / Muller, Philippe / Rivière, Laura / Rutherford, Attapol T. / Zeldes, A mir (2023): "The DISRPT 2023 shared task on elementary discourse unit segmentation, connective detection, and relation classification", in: The Association for Computational Linguistics (ed.): *Proceedings of the 3rd Shared Task on Discourse Relation Parsing and Treebanking (DISRPT 2023)*, Toronto, Canada, 1–21.

Cormen, T homas H. / Leiserson, C harles E. / Rivest, R onald / Stein, Clifford (2009): *Algorithmen – Eine Einführung*. 4. Auflage, Oldenbourg Verlag München.

Dekker, Ronald Haentjens / Hulle, Dirk van / Middell, Gregor / Neyt, Vincent / Zundert, Joris van (2014): "Computer-supported collation of modern manuscripts: CollateX and the Beckett Digital Manuscript Project", in: *Digital Scholarship in the Humanities* 30, 3:452-470. DOI: 10.1093/llc/fqu007.

Dähne, J anis / Ritter, Jörg W. (2024): "Many-to-many sentence alignments", in: Alliance of Digital Humanities Organizations (ed.): *DH2024 Book of Abstracts*, Washington D.C., USA, 141–144. DOI: 10.5281/zenodo.13761066.

Dähne, J anis / Ritter, Jörg W. / Molitor, Paul (2025): "Improving text collations by local resegmentation", in: *Digital Scholarship in the Humanities (DSH)*. DOI: 10.1093/llc/fqaf033.

Habeck, R obert (2002): "Vom Umgang mit Ungeheuern: 'Moby-Dicks' Übersetzer als Walfänger", in: *literaturkritik.de*, no. 5/2002. <https://literaturkritik.de/id/4938> [30.04.2025].

Jaccard, P aul (1901): "Étude comparative de la distribution florale dans une portion des Alpes et du Jura", in: *Bulletin de la Société Vaudoise des Sciences Naturelles* 37, 142:547–579. DOI: 10.5169/SEALS-266450.

Ji, Yangfeng / Eisenstein, Jacob (2014): "Representation Learning for Text-level Discourse Parsing", in: Association for Computational Linguistics (ed.): *Proceedings of the 52nd Annual Meeting*, Baltimore, USA, 1:13–24. DOI: 10.3115/v1/P14-1002.

Knoth, Alexander / Stede, M anfred / Hägert Erik (2018). "Dokumentenarbeit mit hierarchisch strukturierten Texten: Eine historisch vergleichende Analyse von Verfassungen", in: Digital Humanities im deutschsprachigen Raum (ed.): *Konferenzabstracts der DHd2018 Kritik der digitalen Vernunft*, Köln, Germany, 196–201. DOI: 10.5281/zenodo.3684896.

Levenshtein, Vladimir I. (1966): "Binary codes capable of correcting deletions, insertions, and reversals", in: *Soviet Physics Doklady* 10, 8:707–710.

Mann, William C. / Thompson, Sandra A. (1987): *Rhetorical structure theory: A theory of text organization*. University of Southern California, Information Sciences Institute Los Angeles.

Molfese, Francesco / Bejgu, Andrei Stefan / Tedeschi, S imone / Conia, Simone / Navigli Roberto (2024): "CroCoAlign: A cross-lingual, context-aware and fully-neural sentence alignment system for long texts", in: Association for Computational Linguistics (ed.): *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Long Papers)*, 1: 2209–2220.

Pöckelmann, M arcus / Medek, A ndré / Ritter, Jö rg W. / Molitor, Paul (2023): "LERA - An interactive platform for synoptical representations of multiple text witnesses", in: *Digital Scholarship in the Humanities* 38, 1:330–346. DOI: 10.1093/llc/fqac021.

Reimers, Nils / Gurevych, Iryna (2019): "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks", in: Association for Computational Linguistics (ed.): *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, Hong Kong, China, 3982–3992. DOI: 10.18653/v1/D19-1410.

Shahmohammadi, Sara / Stede, M anfred (2024): "Discourse parsing for German with new RST corpora", in: Association for Computational Linguistics (ed.): *Proceedings of the 20th Conference on Natural Language Processing (KONVENS 2024)*, Vienna, Austria, 65–74.

Sidarenka, Uladzimir / Peldszus, Andreas / Stede, M anfred (2015): "Discourse segmentation of German texts", in: *Journal for language technology and Computational Linguistics* 30, 1:71–98.

Stede, M anfred / Neumann, Arne (2014): "Potsdam commentary corpus 2.0: Annotation for discourse research", in: European Language Resources Association (ed.): *Proceedings of the Ninth International International Conference on Language Resources and Evaluation (LREC'14)*, Reykjavik, Iceland, 925–929.

Steingrímmsson, Steinþór / Loftsson, Hrafn / Way, Andy (2023): “SentAlign: Accurate and scalable sentence alignment”, in: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Singapore, 256–263. DOI: 10.18653/v1/2023.emnlp-demo.22.

Thompson, Brian / Koehn, Philipp (2019): “Vecalign: Improved sentence alignment in linear time and space”, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, Hongkong, China, 1342–1348. DOI: 10.18653/v1/D19-1136.

Tofiloski, Milan / Brooke, Julian / Taboada, Maite (2009): “A syntactic and lexical-based discourse segmenter”, in: Association for Computational Linguistics (ed.): *Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP (Short Papers)*, Suntec, Singapore, pages 77–80.

Wang, Yizhong / Li, Sujian / Y ang, Jingfeng (2018): “Toward fast and accurate neural discourse segmentation”, in: Association for Computational Linguistics: *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Brussels, Belgium, 962–967.